Testosterone Levels in two species of blackbirds with different parental and reproductive strategies: Pre- and Post-Stress Analysis

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Introduction

BHCO and COGR Traits

- Brown Head Cow Birds (BHCO)
  - Parasitic (Jones et al., 2023)
  - Small, About 40 eggs per breeding season (Lowther, 2020)
- Common Grackles (COGR)
  - Parental (Audubon, 2014)
  - Large, 4-5 eggs per breeding season (Audubon, 2014).
- COGR is phylogenetically relevant to BHCO
- BHCO and COGR are part of the Icteridae bird family (Cornell Lab of Ornithology, 2024).
- BHCO inhabit similar places to COGR but only parasitize Red Wing Black Bird (RWBB) nests



BHCO COGR

HPA and HPG axis functions and interactions

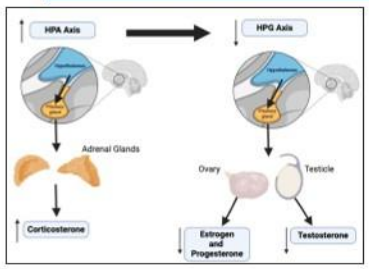


Fig. 1. HPA-HPG interactions. HPA and HPG have known interactions between the stress response and reproductive hormone concentration (Dunlavey, 2018; Acevedo-Rodriguez et al., 2018; Rosvall et al., 2013).

- During times of chronic stress, the HPA axis releases corticosteroids, which impact the HPG axis through the suppression of GnRH; this decreases reproductive hormone production (Ngala Elvis Mbiydzenyuy & Lihle-Appiah Qulu, 2024).
- Both corticosterone and reproductive hormones have many influences on reproduction, parental care, and behaviors in animals. (Lynn, 2016)

Differences in Reproductive hormone responses

- Previous studies by the Lynch Lab have found that BHCO shows abnormal testosterone and estrogen concentration patterns during the breeding season, while COGR follows standard patterns (Fig. 2).
- BHCO shows stable levels of reproductive hormone throughout the breeding season, allowing them to lay many eggs (Dusque et al., 2024)
- Previous data collection by the Lynch Lab found increases in corticosterone when BHCO and COGR underwent captive stress induction (Fig. 3.).

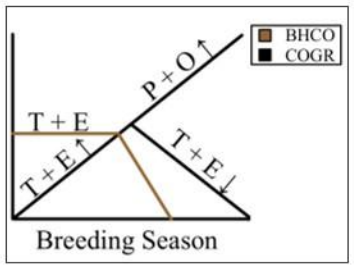


Fig. 2. BHCO and COGR reproductive hormone cycle. T=Testosterone, E=Estrogen, P=Progesterone, O = Oxytocin.

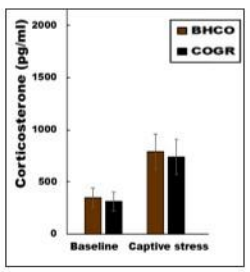


Fig. 3. Increase in Corticosterone Pre and Post Stress induction in BHCO and COGR. Data was collected during the breeding season.

Hypothesis

- Are there differences in how stress impacts the reproductive hormone testosterone within phylogenetically relevant parasitic and parental birds?

Methodology

- **Blood Collection**
  - Trapped and Released within 5 minutes to **minimize stress response**. Blood collected via wing vein
  - Placed in a bander bag for 15 minutes to **induce a stress response**, then a second blood sample was taken.
  - Collected from Morris County, New Jersey and Balcones Wildlife Refuge in Marble Falls, Texas
- **Finding Testosterone Concentration**
  - **Plasma extraction** from the plasma mixtures of BHCO and COGR
  - **Hormone Assay** using the Cayman ELISA Testosterone Kit
    - **Competitive Hormone Assay**: Fluorescence tag particles compete with hormones to bind to pre-attached antibodies. More binding and more fluorescence, thus, lower testosterone concentrations.
  - **Plate Reading** using a BioTek Plate reader at an absorbance of 405 nm
  - **Statistical Analysis** was performed, the mean testosterone concentrations were found, and t-tests were then performed

Results and Discussion I

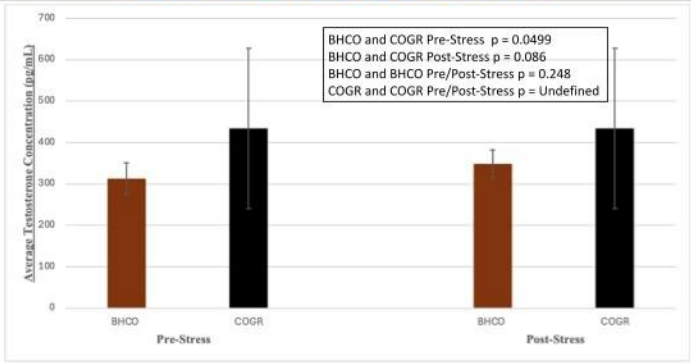


Fig. 3. Average Testosterone Concentration Pre and Post Stress Induction in Brown-headed Cowbird(BHCO) and Common Grackle (COGR)*.

- COGR exhibited **greater observable levels** of testosterone when compared to BHCO pre- and post-stress, respectively (Fig. 3).
  - Birds with higher levels of testosterone are often seen to be more aggressive towards birds with brood parasitism (Abolins-Abols & Hauber, 2018).
  - Host species must adapt to persistent parasitic threats, leading to evolutionary arms races in which increased aggression and hormonal responses become critical survival strategies (Råberg et al., 2008).
  - A common host defense against Brood parasitism is elevated yolk testosterone, which may be present in COGR (Hahn et al., 2017).
- BHCO saw an **observable increase in testosterone concentration** post-stress induction (Fig. 3.)
  - Corticosteroids suppress the HPG axis, inhibiting testosterone (Smith and Franklin, 2021).
  - Corticosteroids have been found to alter aggressive behavior in animals through interactions with testosterone (Ngala Elvis Mbiydzenyuy & Lihle-Appiah Qulu, 2024).

*Figure 3 shows that COGR showed significantly higher testosterone levels than assumed. This caused our dilution factor to be lower than needed, preventing COGR levels from being measured precisely. Therefore, our data does not show a change in COGR rates pre- and post-stress but shows the significant levels of testosterone in COGR.

Results and Discussion II

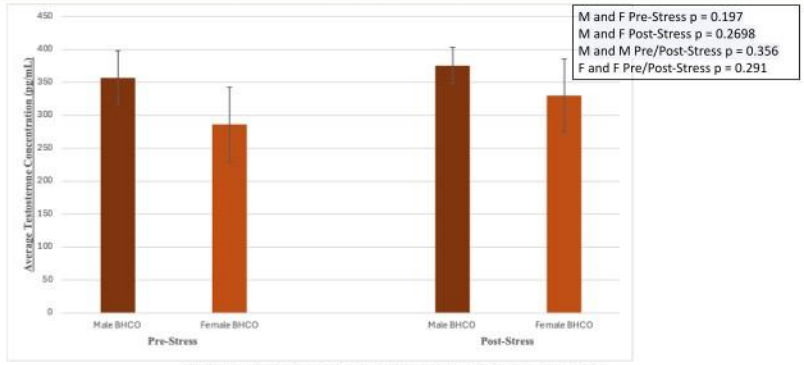


Fig. 4. Male and Female BHCO Pre and Post-Stress Induction.

- Male BHCOs exhibited **observable differences in testosterone** levels pre- and post-stress compared to Female BHCOs (Fig. 4).
- Female BHCO had a **greater observable increase in testosterone** than Male BHCO post-stress (Fig. 4).
  - Testosterone may play a role similar in Females as in Males, with increases in competitive behavior while decreasing paternal and maternal effort (Cain & Ketterson, 2013)
  - Females with higher testosterone often spend less time raising their young (Cain & Ketterson, 2013).

Conclusion/Significance

- COGR has greater observable testosterone pre- and post-stress compared to BHCO
- COGR birds exhibited observable differences in testosterone compared to BHCO pre- and post-stress, respectively, and showed trends toward significance
- Male BHCO have greater testosterone pre- and post-stress compared to Females
- Female BHCO showed a greater increase in testosterone compared to Male BHCO

- Advance our understanding of brood parasitism and the influence of stress on reproductive hormone production and behavior.
  - An increase in testosterone can be related to COGRs' defense against brood parasitism, showing why BHCO do not parasitize them
  - COGR has greater testosterone during peak breeding season than BHCO, a parasitic bird whose levels stay primarily consistent throughout.
  - An increase in HPG axis activation may cause an increase in HPA axis activation

- Further Research
  - Investigate HPG activation impacts on HPA before, during, and after breeding season in parasitic and parental birds

Primary Citations

Abolins-Abols, M., & Hauber, M. E. (2018). Host defences against avian brood parasitism: an endocrine perspective. *Proceedings of the Royal Society B: Biological Sciences*, *285*(1886), 20180980. https://doi.org/10.1098/rspb.2018.0980

Cain, K. E., & Ketterson, E. D. (2013). Individual variation in testosterone and parental care in a female songbird; The dark-eyed junco (Junco hyemalis). *Hormones and Behavior*, *64*(4), 685–692. https://doi.org/10.1016/j.yhbeh.2013.09.002

Duque, F. G., Azam, A., Kaur, A., Pao, R., & Lynch, K. S. (2024). Divergent neural nodes are species- and hormone-dependent in the brood parasitic brain. *Genes, brain, and behavior*, *23*(5), e12907. https://doi.org/10.1111/gbb.12907

Ngala Elvis Mbiydzenyuy, & Lihle-Appiah Qulu. (2024). Stress, hypothalamic-pituitary-adrenal axis, hypothalamic-pituitary-gonadal axis, and aggression. *Metabolic Brain Disease*. https://doi.org/10.1007/s11011-024-01393-w